Use Coding Agents as Research Tools

A Plan-Based Workflow for the Age of Generative AI

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June 2, 2026.
Quantitative and Computational Methods Workshop
Northwestern University

AI is starting to do the research NOW

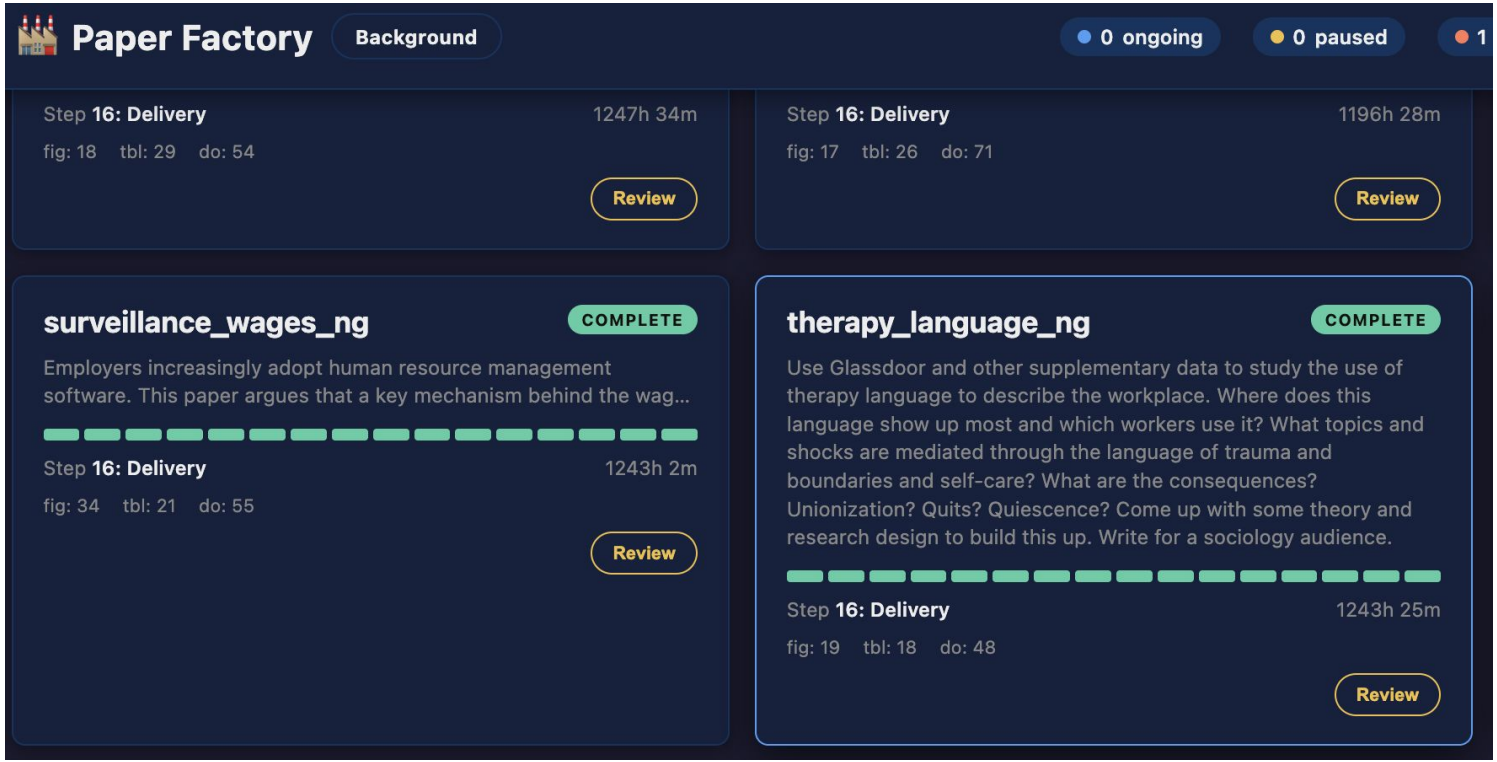
Lu et al (2026), ***The AI Scientist***:

- Generating ideas
- Searching literature
- Planning experiments
- Writing and running code
- Analyzing results
- Plotting figures
- Writing a paper
- Performing review

A conference paper was accepted into the AI workshop (acceptance rate ~ 70%)



Engzell and Wilmers “The Paper Factory”



https://www.nathanwilmers.com/paper_factory/

Prompts

Use Glassdoor and other supplementary data to study the use of therapy language to describe the workplace. Where does this language show up most and which workers use it? What topics and shocks are mediated through the language of trauma and boundaries and self-care? What are the consequences? Unionization? Quits? Quiescence? Come up with some theory and research design to build this up. Write for a sociology audience.

Therapy Language in Employer Reviews: Diffusion and Post-Exit Concentration

Paper Factory

April 11, 2026

Abstract

Workers increasingly describe employers as "toxic," invoke "boundaries," and frame workplace conflict in therapeutic terms. Whether this vocabulary names organizational harm or psychologizes structural problems remains unclear. Using millions of Glassdoor reviews from 2008 to 2022 and a layered within-platform design, this paper shows that therapy language diffused broadly within firms on a major employer-review platform, but the component most closely tied to worker position, the former/current gap within exact firm-year-rating cells, is far narrower than raw comparisons imply. Under the tightest comparison, the remaining residual is concentrated in a single term, "toxic," and appears primarily in the period just after job separation rather than as a generalized wellness idiom. The within-firm rise in therapy language is substantial and not confined to former employees. At the same time, most of the apparent former/current divergence reflects rating severity, and nearly all of the residual is toxicity language deployed shortly after exit and fading within a few years. Employer-review platforms therefore appear to channel displaced voice, with workers deferring challenging speech while employed and directing it to a public audience after leaving.

Keywords: Therapy language | Employer reviews | Exit voice | Toxic workplaces | Glassdoor

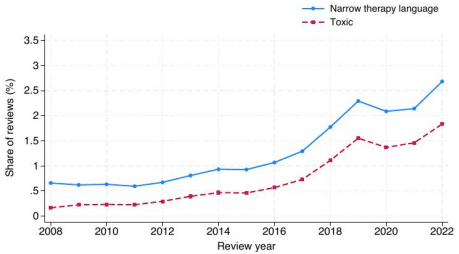


Figure 1. Time Trend in Narrow Therapy Language and Toxic Language, 2008–2022.

Share of Glassdoor reviews containing narrow therapy language (excluding "stress" and "work-life") and "toxic" specifically, by review year. Based on all reviews in the main analysis sample (N = 13,071,350).

drift, or exit endogeneity, and the paper is explicit about these limits throughout. Appendix robustness checks, including reviewer fixed effects, support-threshold restrictions, equal-weight exact-cell benchmarks, and JOLTS sector-year boundary checks, are intended to narrow objections without overclaiming identification.

5. Results

5.1. Therapy language diffused within firms

Figure 1 shows the raw time trend in narrow therapy language and its dominant component, "toxic," on Glassdoor from 2008 to 2022. Both series rise sharply, with therapy language increasing from 0.65% of reviews in 2008 to approximately 2.7% by 2022. "Toxic" accounts for most of that growth, rising from 0.16% to roughly 1.8% over the same period. This increase accelerates after 2016.

How much of this rise occurs within persisting firms, as opposed to reflecting changes in which firms appear on the platform? Table 2 presents the firm-year decomposition. In the pooled model (Column 1), the 2022 coefficient relative to 2012 is 2.818 percentage points, indicating that Glassdoor reviews in 2022 are nearly 3 points more likely to contain therapy language than reviews in the baseline year.

Abstract

Workers increasingly describe employers as “toxic,” invoke “boundaries,” and frame workplace conflict in therapeutic terms. Whether this vocabulary names organizational harm or psychologizes structural problems remains unclear. **Using millions of Glassdoor reviews from 2008 to 2022** and a layered within-platform design, **this paper shows that therapy language diffused broadly within firms** on a major employer-review platform, but the component most closely tied to worker position, the former/current gap within exact firm-year-rating cells, is far narrower than raw comparisons imply. Under the tightest comparison, **the remaining residual is concentrated in a single term, “toxic,” and appears primarily in the period just after job separation rather than as a generalized wellness idiom**. The within-firm rise in therapy language is substantial and not confined to former employees. At the same time, most of the apparent former/current divergence reflects rating severity, and nearly all of the residual is toxicity language deployed shortly after exit and fading within a few years. Employer-review platforms therefore appear to channel displaced voice, with workers deferring challenging speech while employed and directing it to a public audience after leaving.

So, what can *we* do?



But, we are barely using it

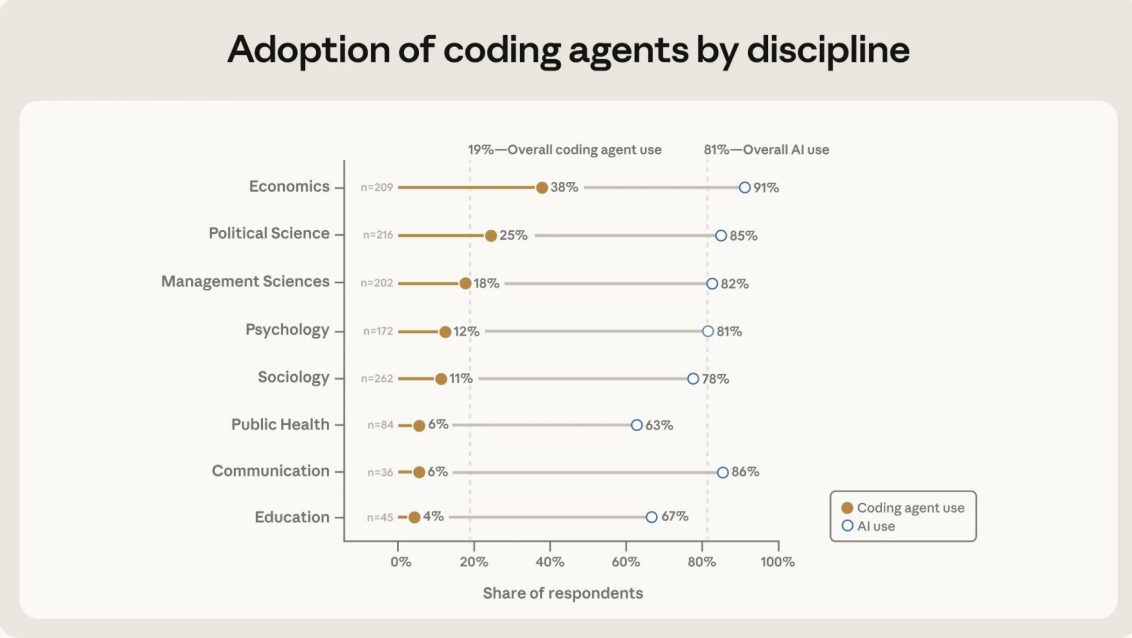


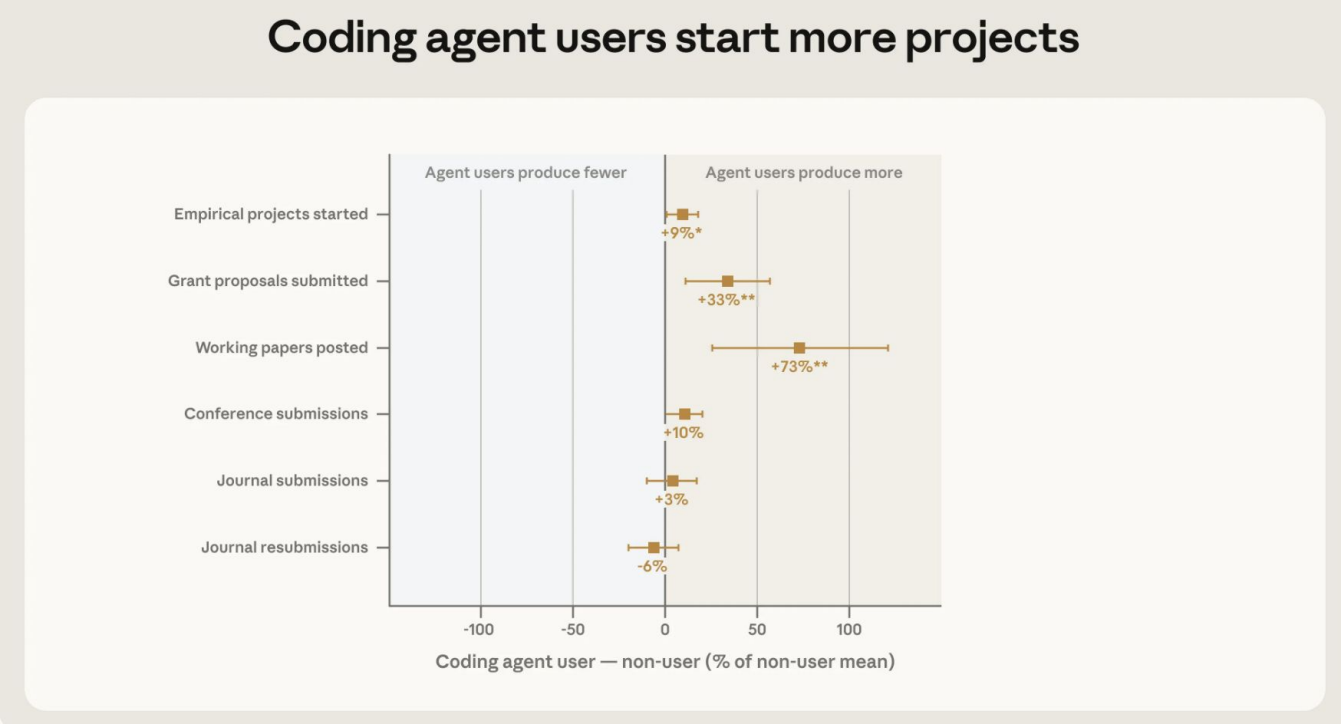
Figure 1: Economists and political scientists report the most use of coding agents. *Coding agent use is yes to "Do you regularly (more than once a week) use an AI coding assistant integrated into your command line (such as Codex, Cursor or Claude Code)?" and a follow-up question verifying use of one of those tools, or Google Antigravity. Discipline categories described in appendix.*

Sociology : 11%

Regularly use a coding agent (codex / cursor / claude code)

Anthropic, "Coding agents in the social science" surveyed 1260 social scientists, Feb-Mar 2026

And the people who use it … start more!



So, why don't sociologists use it?

Type 1: Refuse

- Unease about automation
- Worries about authorship and credit

Type 2: Willing, but stuck

- Want to, but don't know how
- The start-up cost feels high

Type 3: Tried, quit

- Installed it
- hit hallucinations
- Gave up

Today is for **Types 1 and 3**. The **plan-based workflow** answers both at once: it keeps you **accountable** and it makes the agent **reliable**.

From Chatbot (e.g. ChatGPT) **to Agent** (e.g. Codex)

Chatbot

- You ask
- **Chatbot** generate code
- You copy-paste
- You run it
- You debug
- You paste the error back
- Repeat

Agent

- You describe the task
- **Agent** reads your file
- **Agent** writes code
- **Agent** runs it
- **Agent** checkers output
- **Agent** fixes errors
- Repeats until done

The Tireless RA

An RA who can code in any language, works around the clock, and follows your instructions to the letter.

But, it has no independent judgement about your research question.

Your job is direction and verification.

You are the PI. The agent is your RA (PhD level)

The key skill shift from learning to code to **learning to supervise**

Where you can use agents

1. Chatbot / Web interface
2. Codex / Claude app
3. Command Line Interface (CLI) / Terminal
4. Cursor / VS code Editor

Today: **CLI/Terminal within Cursor Editor**

For how to set up: https://github.com/letitbk/claude-academic-setup/

Plan-based workflow

Communicate with the agent (and other collaborators) through a written plan – review it, execute it, and verify against it.

What is plan-based workflow?

Delegation (auto research)

Agent decides, you review

- Agent makes the research decisions
- You inspect the output at the end
- Judgement is late

Plan-based (semi-auto)

You decide, agents executes

- **Write the plan as a markdown file**
- **Review it with agent / coauthor, then revise**
- Agent executes; you verify vs the plan

No plan (zero automation)

You decide, agents executes

- You make the research direction
- Agents executes
- You verify

The loop, step by step



Main workflow

1. **Plan** : tell the agent what and why (not how). Review its plan. Revise.
2. **Execute**: Agent writes code, runs it, checks output autonomously.
3. **Validate**: Verify results, get second opinions, create audit trail.

You can use it for everything (coding, literature review, writing, searching, … etc), big or small.

Why plan? Four payoffs

1. **Portable**: Any model/agent can read a plan (e.g., plan with an expensive model and execute with a cheap one)
2. **Shared**: Coauthors read & review the same document
3. **Verifiable**: The plan is the specification you check the results against.
4. **Ownership**: You set the questions and the standards. Anyone (including yourself or other RAs) can do. You can use the agent as **as a tool** to complete the task.

Agent can make the work faster, but cannot be accountable for the claim.

Tools (claude code plugins) that support the loop

Make it ask before acting	`AskUserQuestion`	forces the choices into the open
Brainstorm options	`Superpowers`	richer ideation; ends in your decision
Review the plan visually	`plannotator`	annotate the plan & diffs, send back
Review claims with coauthors	`discuss-claims`	one GitHub issue per empirical claim
Don't make it forget	`CLAUDE.md · /napkin`	project memory across sessions

Also see: https://github.com/letitbk/claude-academic-setup/

What I could do with plan-based workflows

1. Add “LLM chatbots” into Qualtrics Survey
2. Create a data annotation coding web platform
3. Collect ~10,000 sociology PhD graduates’ careers (from first job to current job) through online search (CV/webpage/linkedin…) + LLM annotation
4. Design a survey scale with autonomous agent loops
5. Fine-tune LLMs on the NYU HPC cluster
6. Using an unfamiliar API for data collection at scale
7. ….

Live Demo

Analysis of GSS data: “Predictability of partisanship”
Analysis of text data: What do American people worry about? (from dear Abby)

GSS data analysis: predictability of partisanship

Prompt: I want to use GSS data (see @data folder) to examine how predictable respondents' party identification is from their demographics and issue opinions, framing this predictability as a practical measure of partisan sorting that helps us understand polarization over time. I am interested in which opinions or demographic characteristics drive rising polarization/sorting. First **investigate how this question could be approached from multiple perspectives**. Then **ask me clarifying questions in multiple rounds**. After I answer, **write a detailed analysis plan** for an R/Stata/Python workflow that examines the data creatively and rigorously, what verification checks are needed, and what report tables, figures should be produced.

Text analysis of “Dear Abby” question data

Prompt: I want to use the Dear Abby question data I downloaded from The Pudding (see @data folder) to study how people’s everyday concerns change over time, and whether external political or social events shape the kinds of problems people bring to Abby; my broader motivation is to connect private-life anxieties to polarization, political conflict, and changing public events. **First explore this question from multiple perspectives**. Then **ask me clarifying questions in multiple rounds**. After I answer, **write a detailed plan for** an R/Stata/Python workflow that examines the data creatively and rigorously, what verification checks are needed, and what report tables, figures should be produced.

Why your first attempts may fail

Three things about how LLMs work

It predicts next token from everything in the context window

Why your attempt may fail?

- LLM predicts next token given **the whole context window** through conditional probability.
- It leads to the **context contamination**

What to do differently

- Start a fresh session. Use /compact or /clear the context often.
- Put a durable rule in CLAUDE.md or AGENTS.md (or napkin.md)

Long contexts forget

Why your attempt may fail?

- Even with big windows, models attend unevenly (e.g., **the middle gets lost**), and long sessions drift past the rule you gave 40 messages ago.
- A decoder-only model : **It doesn’t look back** automatically (unless we ask)

What to do differently

- Keep plans short and re-readable.
- Give it a second chance with fresh sessions.
- Use new agents (or other agents) to review the plan.

Two levels: skill and subagents

Skills for consistency

- When **you do the same task** the sameway repeatedly (e.g., cleaning survey data following certain rules, checking data after merge, using the same API, ….)
- Once set up, agents can deploy the relevant skills for relevant tasks automatically.

Sub-agents for isolation

- **Subagent has its own clean context**
- You can parallelize tasks without contaminating your main agents' "context windows"

How to plan well

1. Ask **“what” and “why”** — not “how”
2. Make agents **ask you the questions back** (e.g., *AskUserQuestion* tool)
3. Ask for a “detailed” plan
4. Review, review, review (e.g., *plannotator plugin*)
5. Cross-review with multiple agents (e.g. codex, gemini, claude, ….)

Use plan mode (SHIFT+TAB) for any non-trivial task.

You don’t have to use plan mode for everything though.

How to get started

1. Install codex / claude code / gemini agents
2. (optional but highly recommend) Install VS code or Cursor
3. Open Terminal
    a. Cursor / VS code: using CMD+SHIFT+P > Terminal: Create New Terminal
    b. Mac finder : Right click to the target folder > Services > New Terminal at Folder
    c. Or type “terminal” in the spotlight search (CMD + SPACE)
4. In the terminal, type “claude --dangerously-skip-permissions” or “codex –yolo”
5. Ask agents
    a. “I would like to install the plugin and skill from this repo: https://github.com/letitbk/claude-academic-setup/”
6. And then start your journey!

Q&A



FAQ?

https://github.com/letitbk/claude-academic-setup#faq

1. Do I need to make a plan always?
2. Which model / plan should I use: Codex vs Claude Code vs Gemini
3. What about sensitive / restricted data?
4. Can I safely use “--yolo” or “--dangerously-skip-permissions” mode?
5. Is it better to “/resume” the conversation or /compact or /clear? : prompt caching
6. ….

The skill is learnable – and the ceiling is high

When I taught my PhD students this workflow, they ended up using it better than I do!
Don’t begin with automation. **Begin with a plan.**